

Shaft Component Generator (Version: 2021 (Build 250183000, 183))

03/11/2021

Project Info

Calculation

Material

Material		User material
Modulus of Elasticity	E	206000 MPa
Modulus of Rigidity	G	80000 MPa
Density	ρ	7860 kg/m^3

Calculation Properties

Include		
No	Density	ρ 7860 kg/m^3
Yes	Shear Displacement Ratio	β 1.188 ul
	Number of Divisions	1000 ul
	Mode of reduced stress	HMH

Loads

Index	Location	Radial Force				Bending Moment				Continuous Load				Axial Force	Torque	Deflection				Deflection Angle
		Y	X	Size	Direction	Y	X	Size	Direction	Y	X	Size	Direction			Y	X	Size	Direction	
1	53 mm	981.000 N	-327.000 N	1034.065 N	341.57 deg										-32.500 N m	89.669 microm	-25.958 microm	93.351 microm	343.86 deg	0.13 deg
2	190 mm	-490.000 N	120.000 N	504.480 N	166.24 deg											362.741 microm	-103.257 microm	377.151 microm	344.11 deg	0.09 deg
3	600 mm		571.200 N	571.200 N	90.00 deg										65.000 N m	836.991 microm	-213.457 microm	863.781 microm	345.69 deg	0.05 deg
4	785 mm	-3434.000 N	120.000 N	3436.096 N	178.00 deg											726.438 microm	-165.759 microm	745.110 microm	347.15 deg	0.17 deg
5	930 mm	2943.000 N	-484.000 N	2982.533 N	350.66 deg										-32.500 N m	92.832 microm	-20.322 microm	95.031 microm	347.65 deg	0.31 deg

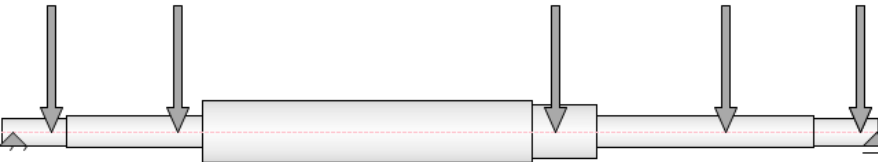
Supports

Index	Type	Location	Reaction Force				Axial Force	Yielding	Type	Deflection				Deflection Angle
			Y	X	Size	Direction				Y	X	Size	Direction	
1	Fixed	12 mm	-0.206 N	8.462 N	8.465 N	91.40 deg			User	0.000 microm	-0.000 microm	0.000 microm	344.05 deg	0.13 deg
2	Free	948 mm	0.206 N	-8.262 N	8.265 N	271.43 deg			User	0.000 microm	-0.000 microm	0.000 microm	335.40 deg	0.30 deg

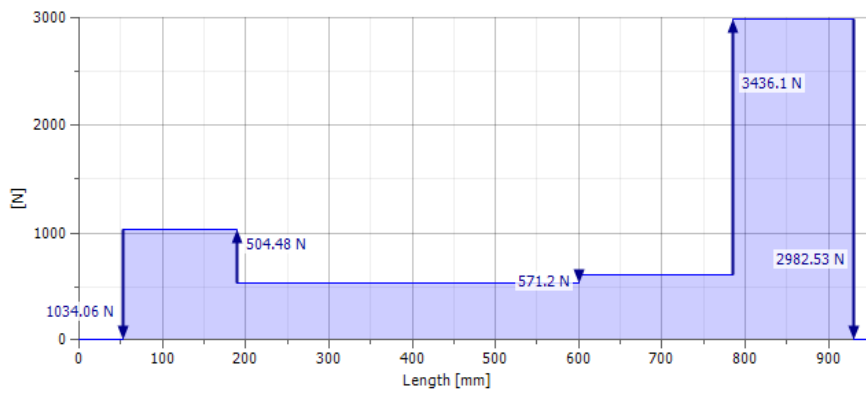
Results

Length	L	950.000 mm
Mass	Mass	15.290 kg
Maximal Bending Stress	σ _B	102.683 MPa
Maximal Shear Stress	τ _S	4.217 MPa
Maximal Torsional Stress	τ	6.130 MPa
Maximal Tension Stress	σ _T	0.000 MPa
Maximal Reduced Stress	σ _{red}	103.034 MPa
Maximal Deflection	f _{max}	909.573 microm
Angle of Twist	φ	0.02 deg

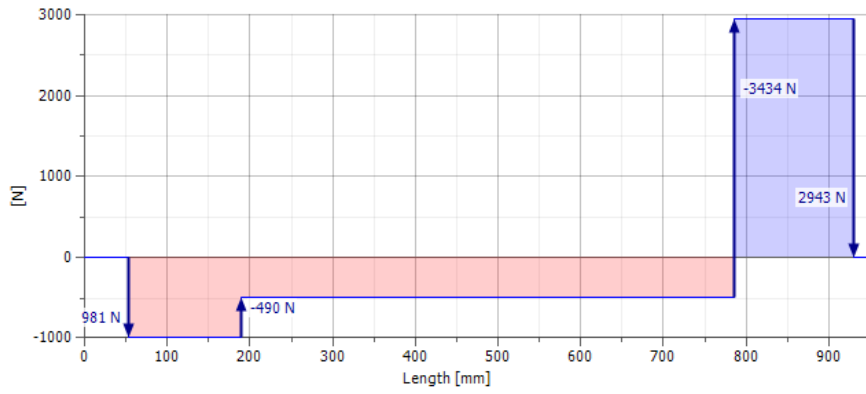
Preview



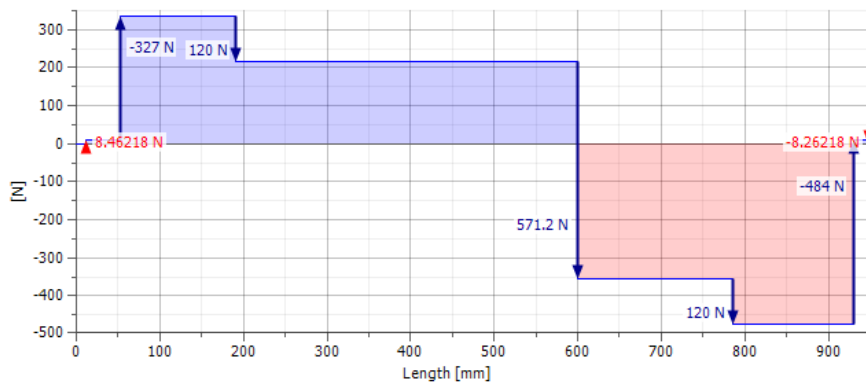
Shear Force



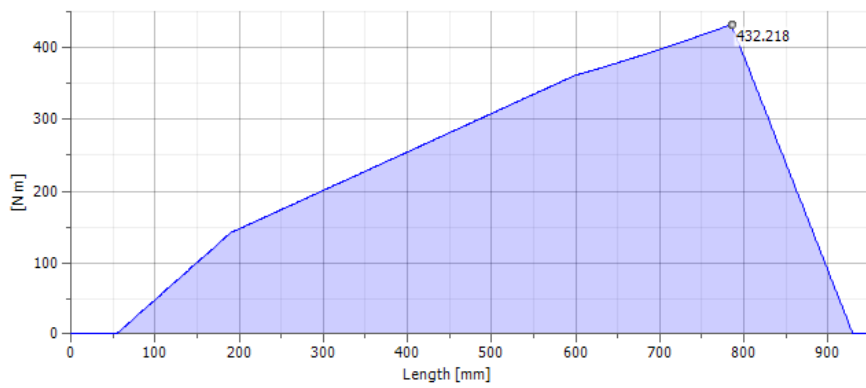
Shear Force, YZ Plane



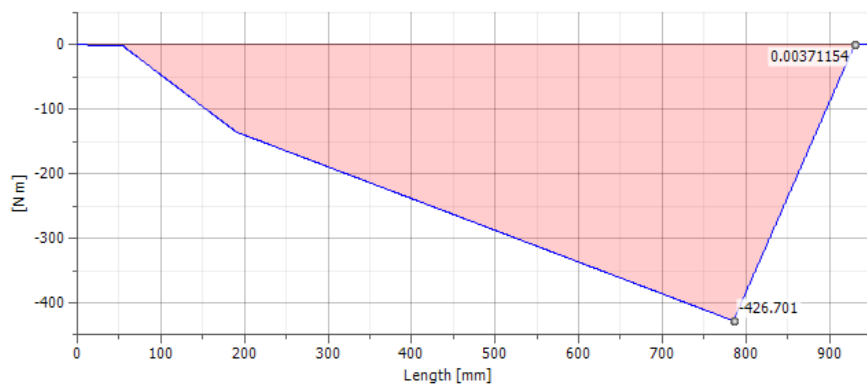
Shear Force, XZ Plane



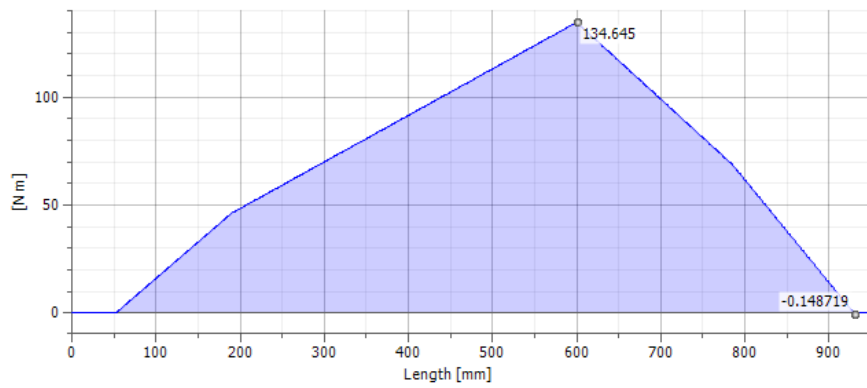
Bending Moment



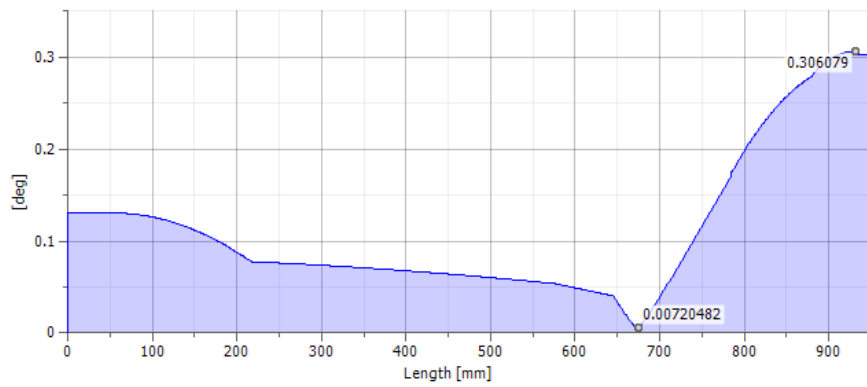
Bending Moment, YZ Plane



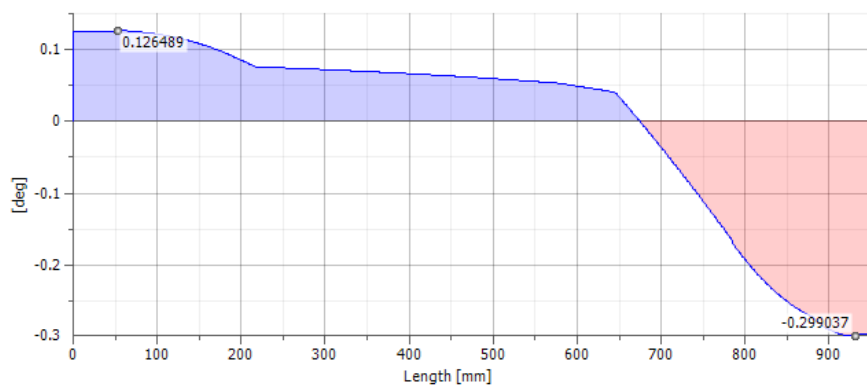
Bending Moment, XZ Plane



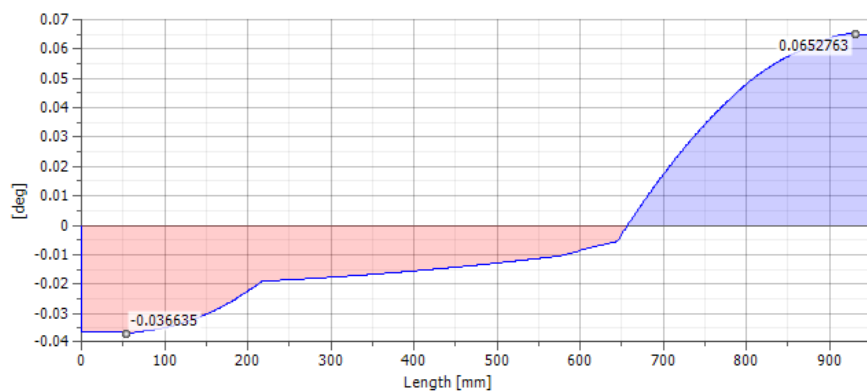
Deflection Angle



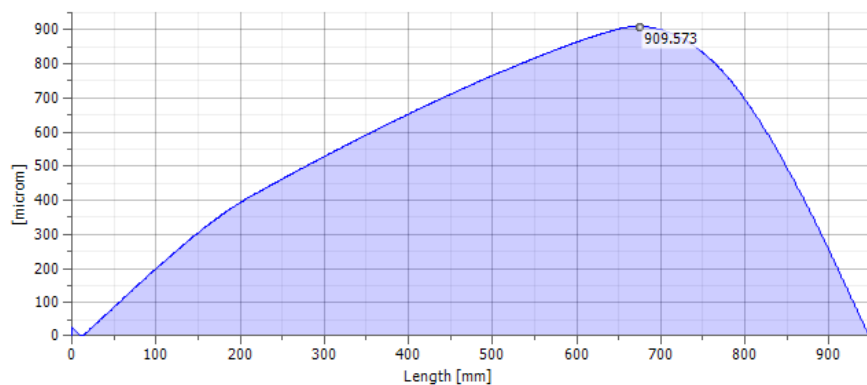
Deflection Angle, YZ Plane



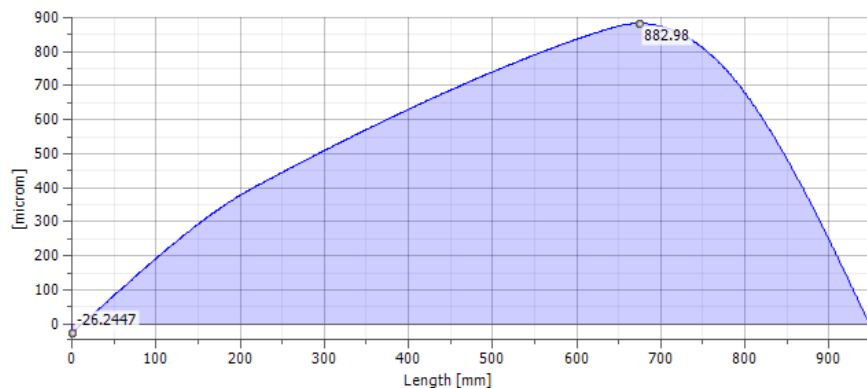
Deflection Angle, XZ Plane



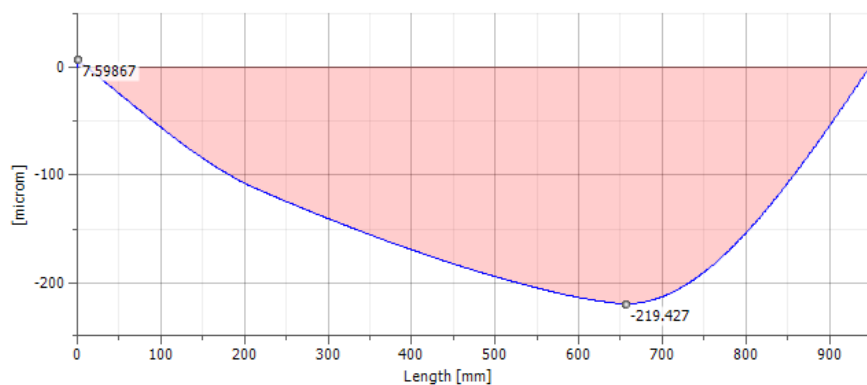
Deflection



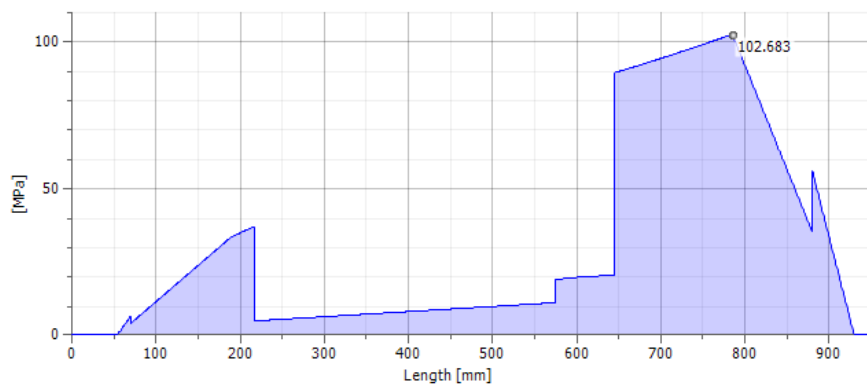
Deflection, YZ Plane



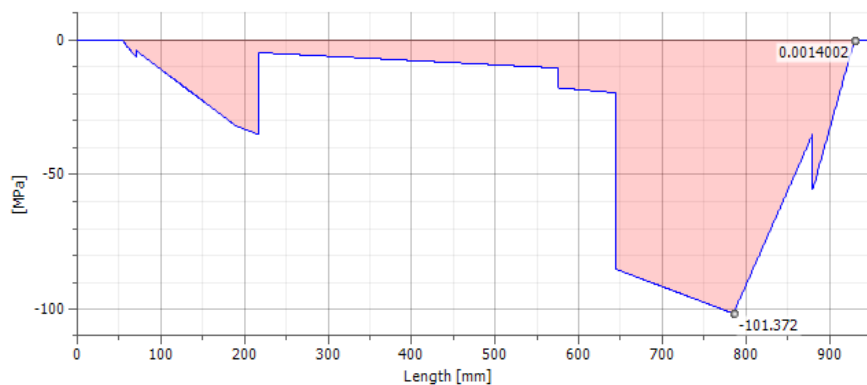
Deflection, XZ Plane



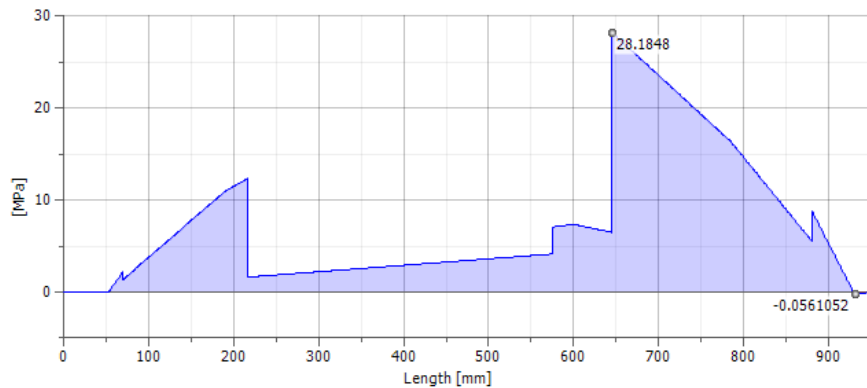
Bending Stress



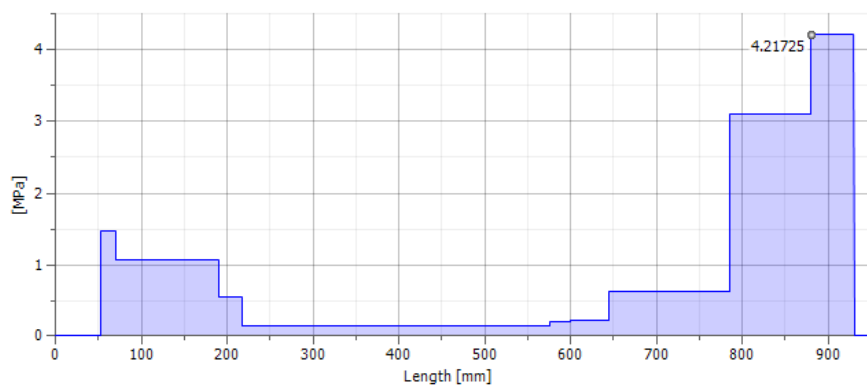
Bending Stress, YZ Plane



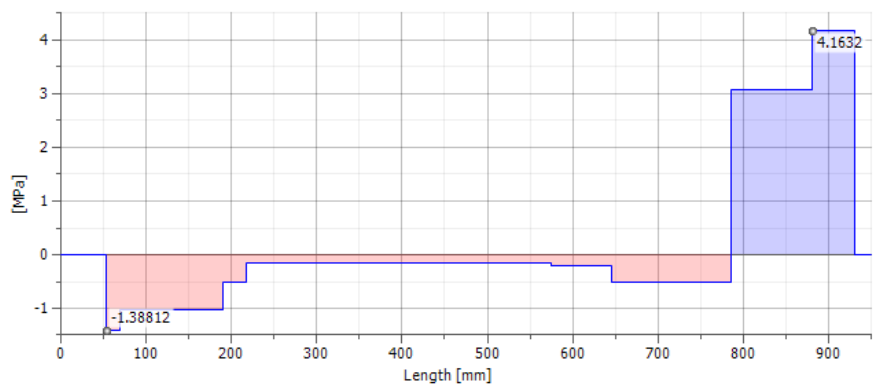
Bending Stress, XZ Plane



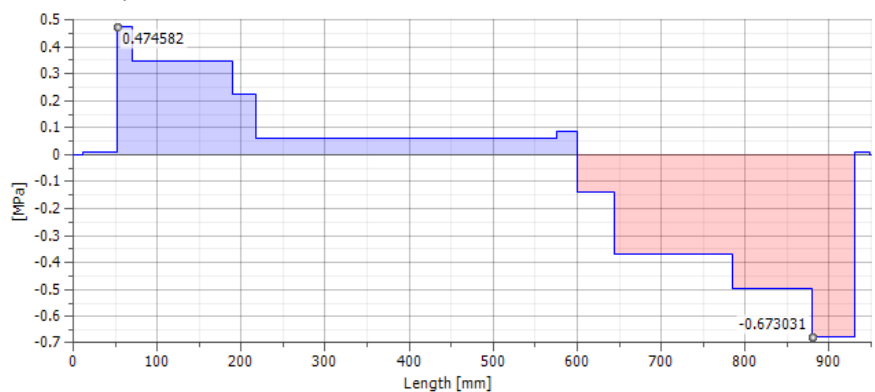
Shear Stress



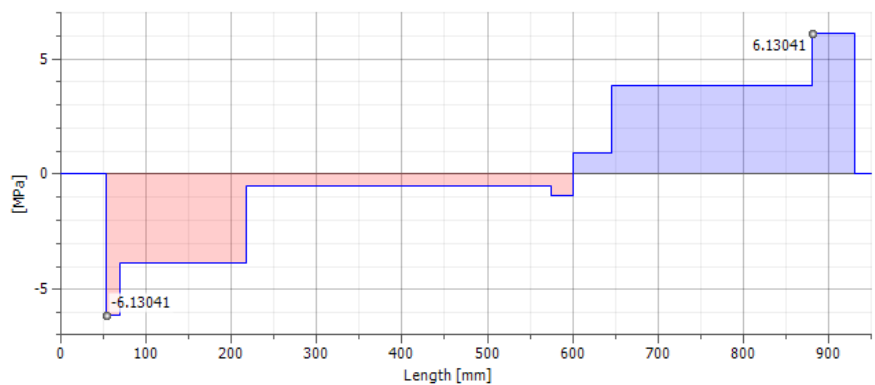
Shear Stress, YZ Plane



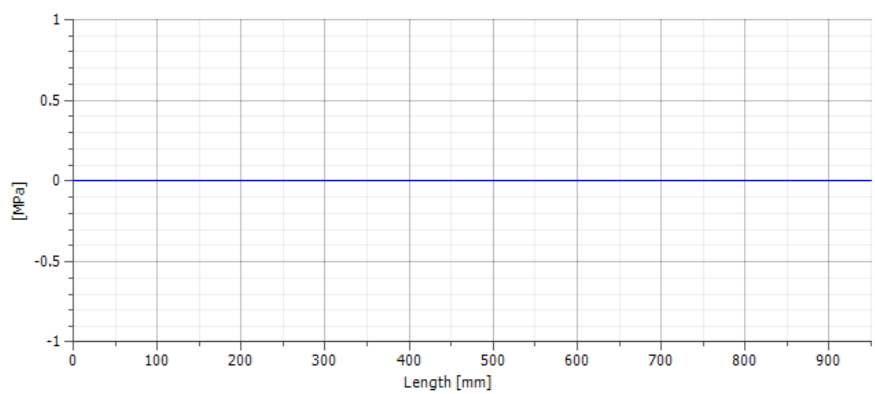
Shear Stress, XZ Plane



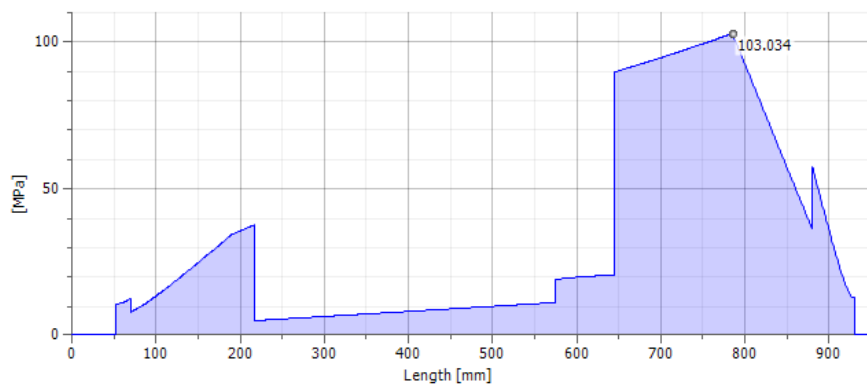
Torsional Stress



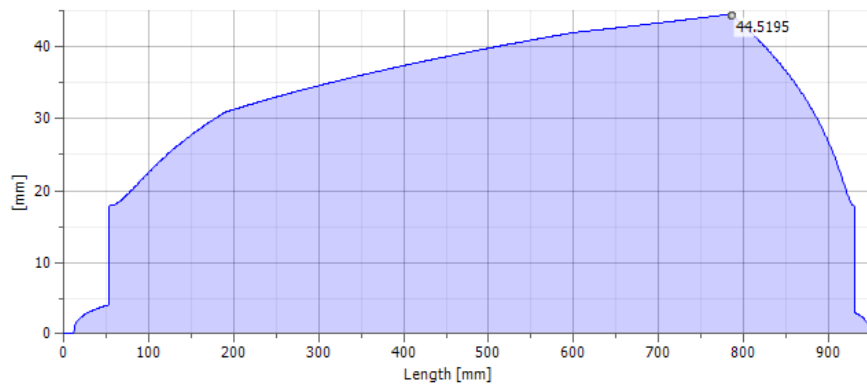
Tension Stress



Reduced Stress



Ideal Diameter



Summary of Messages

02:33:40 Calculation: Calculated.